

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: CRYPTOGRAPHY AND NETWORK
SECURITY COURSE CODE: CSA5117**

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Daniel Jesuraji J(192565085) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Annexure A Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

A. Explain why this issue occurs.

A company is transmitting repeated financial data such as account numbers, transaction IDs, and payment records over a network. Initially, the company uses **Electronic Codebook (ECB)** mode for encryption. In ECB mode, each plaintext block is encrypted independently using the same secret key. As a

result, if two plaintext blocks contain identical information, they produce exactly the same ciphertext block.

For example, if the plaintext contains repeated blocks like:

P1 = Salary
P2 = Salary
P3 = Transfer
P4 = Salary

The corresponding ciphertext in ECB mode will also contain repeated encrypted blocks. Although the actual content is hidden, an attacker observing the encrypted data can easily recognize repeated patterns. This information can help an attacker analyze the communication and make assumptions about the original data. Therefore, ECB mode does not provide semantic security and is not suitable for protecting confidential financial information.

B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

A better choice is **Cipher Block Chaining (CBC)** mode.

In CBC mode, before encrypting each plaintext block, it is first XORed with the previous ciphertext block. The first plaintext block is XORed with an **Initialization Vector (IV)**. Since each block depends on the previous ciphertext, even if two plaintext blocks are identical, their ciphertext blocks will be completely different.

Advantages of CBC mode:

- Prevents repeated ciphertext patterns.
- Provides better confidentiality for financial transactions.
- Makes cryptanalysis more difficult.
- Suitable for transmitting banking and business data over public networks.
- Improves overall security compared to ECB.

Conclusion:

For transmitting sensitive financial records, CBC mode is recommended because it hides plaintext patterns and offers stronger protection against attackers.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

A. Identify the most suitable algorithm for this scenario.

Among DES, 3-DES, and Blowfish, the **most suitable algorithm is Blowfish**.

B. Justify your choice based on security strength and efficiency.

A banking application requires both high security and good performance because thousands of customer transactions are processed every second. The encryption algorithm must protect confidential information while minimizing processing delays.

DES (Data Encryption Standard)

- Key size: 56 bits
- Encryption speed: Moderate
- Security: Weak by today's standards
- Vulnerable to brute-force attacks
- Not recommended for banking applications

3-DES (Triple DES)

- Key size: 168 bits
- Encrypts data three times using DES
- Much stronger than DES
- Very slow because encryption is performed three times
- High computational cost

Blowfish

- Variable key size from 32 to 448 bits
- Very fast encryption
- Excellent security
- Resistant to brute-force attacks
- Low memory requirement
- Suitable for large amounts of banking data

Comparison

Algorithm	Security	Speed	Suitable for Banking
DES	Low	Medium	No
3-DES	Very High	Slow	Moderate
Blowfish	High	Fast	Yes

Conclusion:

Blowfish is the best choice because it provides strong encryption with excellent performance. It balances security and efficiency, making it ideal for banking applications where large numbers of secure transactions are processed continuously.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

A. Explain how the key exchange works in this scenario.

Diffie–Hellman Key Exchange enables two users to establish a common secret key over an insecure public network without transmitting the secret key itself.

The process is as follows:

1. Both users publicly agree on a prime number (p) and a generator (g).
2. User A selects a private key and calculates a public key using modular exponentiation.
3. User B also selects a private key and calculates another public key.
4. The public keys are exchanged over the public network.
5. User A uses User B's public key along with A's private key to compute the shared secret.
6. User B uses User A's public key along with B's private key to compute the same shared secret.
7. Since both users obtain the same secret key, they can use it for secure encryption and decryption.

Although the communication channel is public, the actual secret key is never transmitted, making the exchange secure against passive eavesdropping.

B. Identify a possible security threat and suggest a solution to mitigate it.

The biggest threat to Diffie–Hellman is the **Man-in-the-Middle (MITM) attack**.

During the exchange, an attacker intercepts the public keys sent by both users. Instead of forwarding the original public keys, the attacker sends fake public keys to each user. As a result, User A establishes a secret key with the attacker, and User B also establishes another secret key with the attacker.

The attacker can decrypt messages received from one user, read or modify them, encrypt them again, and send them to the other user without either user realizing the attack.

Solution

The attack can be prevented by authenticating the exchanged public keys.

Methods include:

- Digital Certificates
- Public Key Infrastructure (PKI)

- Digital Signatures
- TLS/SSL protocols
- Authenticated Diffie–Hellman

These methods verify the identity of communicating users and ensure that the exchanged public keys are genuine.

Conclusion:

Diffie–Hellman provides secure key exchange, but authentication mechanisms must be used to prevent MITM attacks.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

A. Describe how RSA can be used for encryption and decryption in this scenario.

RSA is an asymmetric cryptographic algorithm that uses two different keys:

- Public Key
- Private Key

When a sender wants to send a confidential email:

1. The sender obtains the receiver's public key.
2. The email is encrypted using the receiver's public key.
3. The encrypted email is transmitted over the Internet.
4. Only the receiver possesses the corresponding private key.
5. The receiver decrypts the email using the private key and reads the original message.

Because only the intended receiver owns the private key, unauthorized users cannot decrypt the email.

RSA can also be used for digital signatures. The sender signs the email using their private key, and the receiver verifies the signature using the sender's public key. This guarantees message authenticity, integrity, and non-repudiation.

B. Explain how key distribution and management play a role in maintaining security.

The security of RSA depends not only on encryption but also on proper key management.

Important aspects include:

- Secure generation of public and private keys.
- Safe storage of private keys.
- Distribution of authentic public keys using Digital Certificates.
- Regular key renewal.
- Revoking compromised keys.
- Maintaining Certificate Authorities (CA) to verify identities.

Without proper key management, attackers could replace public keys with fake ones and intercept confidential communications.

Therefore, effective key management is essential for maintaining secure email communication.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.
- B. Recommend a suitable cryptographic approach and justify your answer.

A. Analyze the constraints of IoT devices.

IoT devices such as smart sensors, smart home appliances, wearable devices, and industrial monitoring systems have limited hardware resources.

The major constraints include:

- Low processing power
- Limited RAM
- Small storage capacity
- Limited battery life
- Low network bandwidth
- Limited computational capability
- Need for real-time communication
- Low-cost hardware

Because of these limitations, traditional cryptographic algorithms requiring large key sizes and high computational power may reduce device performance and battery life.

B. Recommend a suitable cryptographic approach and justify your answer.

The most suitable cryptographic technique for IoT devices is **Elliptic Curve Cryptography (ECC)**.

ECC provides the same level of security as RSA while using much smaller key sizes.

For example:

- RSA requires approximately **1024-bit** keys.
- ECC provides equivalent security with only **160-bit** keys.

Because of its smaller key size, ECC:

- Performs encryption faster.
- Requires fewer CPU cycles.
- Uses less memory.
- Consumes less battery power.
- Reduces communication overhead.
- Improves device lifetime.
- Maintains strong security against attacks.

Comparison

Feature	RSA	ECC
Key Size	Large	Small
Processing Speed	Slower	Faster
Memory Usage	High	Low
Power Consumption	High	Low
Suitable for IoT	No	Yes
Security	High	High

Conclusion

ECC is the best cryptographic solution for IoT devices because it provides strong security with low computational cost, low memory usage, and low power consumption. It is widely used in modern smart home systems, wearable devices, wireless sensor networks, and other resource-constrained environments.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well reasoned, logical decisions	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided

		with strong rationale		n	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand